

The objective of this notebook is to load the house prediction dataset into MongoDB and make queries.

Steps:

Create a Database and a Collection

I created a new database and collection in MongoDB with the house price prediction data (see pdf images file)

Install and import necessary tools

```
In [16]: # already installed
# pip install pymongo
from pymongo import MongoClient
import pandas as pd
```

Connect to MongoDB

```
In [17]: client = MongoClient('mongodb://localhost:27017/')
```

Select the database House_Price_Prediction_db

```
In [18]: db = client['House_Price_Prediction_db']
```

Select the collection House_data

```
In [19]: collection = db['House_data']
```

Retrieve the first 4 records

```
In [20]: # Retrieve the first four records (the first four documents)
result = collection.find().limit(4)

# print the records
for record in result:
    print(record)
```

```
{ '_id': ObjectId('6a256bbf66570f071a28fdd7'), 'id': 7129300520, 'date': '20141013T00
0000', 'price': 221900, 'bedrooms': 3, 'bathrooms': 1, 'sqft_living': 1180, 'sqft_lo
t': 5650, 'floors': 1, 'waterfront': 0, 'view': 0, 'condition': 3, 'grade': 7, 'sqft
_above': 1180, 'sqft_basement': 0, 'yr_built': 1955, 'yr_renovated': 0, 'zipcode': 9
8178, 'lat': 47.5112, 'long': -122.257, 'sqft_living15': 1340, 'sqft_lot15': 5650}
{ '_id': ObjectId('6a256bbf66570f071a28fdd8'), 'id': 6414100192, 'date': '20141209T00
0000', 'price': 538000, 'bedrooms': 3, 'bathrooms': 2.25, 'sqft_living': 2570, 'sqft
_lot': 7242, 'floors': 2, 'waterfront': 0, 'view': 0, 'condition': 3, 'grade': 7, 's
qft_above': 2170, 'sqft_basement': 400, 'yr_built': 1951, 'yr_renovated': 1991, 'zip
code': 98125, 'lat': 47.721, 'long': -122.319, 'sqft_living15': 1690, 'sqft_lot15':
7639}
{ '_id': ObjectId('6a256bbf66570f071a28fdd9'), 'id': 5631500400, 'date': '20150225T00
0000', 'price': 180000, 'bedrooms': 2, 'bathrooms': 1, 'sqft_living': 770, 'sqft_lo
t': 10000, 'floors': 1, 'waterfront': 0, 'view': 0, 'condition': 3, 'grade': 6, 'sqf
t_above': 770, 'sqft_basement': 0, 'yr_built': 1933, 'yr_renovated': 0, 'zipcode': 9
8028, 'lat': 47.7379, 'long': -122.233, 'sqft_living15': 2720, 'sqft_lot15': 8062}
{ '_id': ObjectId('6a256bbf66570f071a28fdda'), 'id': 2487200875, 'date': '20141209T00
0000', 'price': 604000, 'bedrooms': 4, 'bathrooms': 3, 'sqft_living': 1960, 'sqft_lo
t': 5000, 'floors': 1, 'waterfront': 0, 'view': 0, 'condition': 5, 'grade': 7, 'sqft
_above': 1050, 'sqft_basement': 910, 'yr_built': 1965, 'yr_renovated': 0, 'zipcode':
98136, 'lat': 47.5208, 'long': -122.393, 'sqft_living15': 1360, 'sqft_lot15': 5000}
```

Query: Find the document with id 1175000570

```
In [21]: # Find one document:
# define the query filter
query = {'id':1175000570}

# Retrieve the document with id 1175000570
result = collection.find_one(query)
print(result)
```

```
{ '_id': ObjectId('6a256bbf66570f071a28fde5'), 'id': 1175000570, 'date': '20150312T00
0000', 'price': 530000, 'bedrooms': 5, 'bathrooms': 2, 'sqft_living': 1810, 'sqft_lo
t': 4850, 'floors': 1.5, 'waterfront': 0, 'view': 0, 'condition': 3, 'grade': 7, 'sq
ft_above': 1810, 'sqft_basement': 0, 'yr_built': 1900, 'yr_renovated': 0, 'zipcode':
98107, 'lat': 47.67, 'long': -122.394, 'sqft_living15': 1360, 'sqft_lot15': 4850}
```

Create a dataframe using all the records from the collection

```
In [25]: # Retrieve all records
result = collection.find({})

# Convert the result to a pandas Dataframe
df = pd.DataFrame(list(result))
print(df)
```

	_id	id	date	price	\
0	6a256bbf66570f071a28fdd7	7129300520	20141013T000000	221900.0	
1	6a256bbf66570f071a28fdd8	6414100192	20141209T000000	538000.0	
2	6a256bbf66570f071a28fdd9	5631500400	20150225T000000	180000.0	
3	6a256bbf66570f071a28fdda	2487200875	20141209T000000	604000.0	
4	6a256bbf66570f071a28fddb	1954400510	20150218T000000	510000.0	
...	...	...	...	...	
21608	6a256bc066570f071a29523f	263000018	20140521T000000	360000.0	
21609	6a256bc066570f071a295240	6600060120	20150223T000000	400000.0	
21610	6a256bc066570f071a295241	1523300141	20140623T000000	402101.0	
21611	6a256bc066570f071a295242	291310100	20150116T000000	400000.0	
21612	6a256bc066570f071a295243	1523300157	20141015T000000	325000.0	

	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	...	\
0	3	1.00	1180	5650	1.0	0	...	
1	3	2.25	2570	7242	2.0	0	...	
2	2	1.00	770	10000	1.0	0	...	
3	4	3.00	1960	5000	1.0	0	...	
4	3	2.00	1680	8080	1.0	0	...	
...	...	...	...	...	...	...	...	
21608	3	2.50	1530	1131	3.0	0	...	
21609	4	2.50	2310	5813	2.0	0	...	
21610	2	0.75	1020	1350	2.0	0	...	
21611	3	2.50	1600	2388	2.0	0	...	
21612	2	0.75	1020	1076	2.0	0	...	

	grade	sqft_above	sqft_basement	yr_built	yr_renovated	zipcode	\
0	7	1180	0	1955	0	98178	
1	7	2170	400	1951	1991	98125	
2	6	770	0	1933	0	98028	
3	7	1050	910	1965	0	98136	
4	8	1680	0	1987	0	98074	
...	...	...	...	...	...	...	
21608	8	1530	0	2009	0	98103	
21609	8	2310	0	2014	0	98146	
21610	7	1020	0	2009	0	98144	
21611	8	1600	0	2004	0	98027	
21612	7	1020	0	2008	0	98144	

	lat	long	sqft_living15	sqft_lot15
0	47.5112	-122.257	1340	5650
1	47.7210	-122.319	1690	7639
2	47.7379	-122.233	2720	8062
3	47.5208	-122.393	1360	5000
4	47.6168	-122.045	1800	7503
...	...	...	...	...
21608	47.6993	-122.346	1530	1509
21609	47.5107	-122.362	1830	7200
21610	47.5944	-122.299	1020	2007
21611	47.5345	-122.069	1410	1287
21612	47.5941	-122.299	1020	1357

[21613 rows x 22 columns]

Display the first five records of the DataFrame

In [26]: `df.head(5)`

Out[26]:

	_id	id	date	price	bedrooms	bathroom
0	6a256bbf66570f071a28fdd7	7129300520	20141013T000000	221900.0	3	1.0
1	6a256bbf66570f071a28fdd8	6414100192	20141209T000000	538000.0	3	2.0
2	6a256bbf66570f071a28fdd9	5631500400	20150225T000000	180000.0	2	1.0
3	6a256bbf66570f071a28fdda	2487200875	20141209T000000	604000.0	4	3.0
4	6a256bbf66570f071a28fddb	1954400510	20150218T000000	510000.0	3	2.0

5 rows × 22 columns



Display the name of all the columns in the DataFrame

In [27]: `df.columns`

Out[27]: Index(['_id', 'id', 'date', 'price', 'bedrooms', 'bathrooms', 'sqft_living', 'sqft_lot', 'floors', 'waterfront', 'view', 'condition', 'grade', 'sqft_above', 'sqft_basement', 'yr_built', 'yr_renovated', 'zipcode', 'lat', 'long', 'sqft_living15', 'sqft_lot15'], dtype='object')

Credits: This relies on the material in Professor Angel Martinon Hernandez's CISD 43 class at Mt. San Antonio College, Spring semester 2026, including instructor code.